

Editorial

Soil and human health

Soil can affect human health in many ways, either directly or indirectly, although the causal relationship can be difficult to establish (Oliver, 1997). Soil provides essential nutrients through plants to animals and humans; inadequate supply of these nutrients in soil can cause deficiencies in humans, notable examples being selenium and zinc. Soil may contain elevated amounts of toxic elements naturally, or be contaminated with inorganic or organic contaminants by human activities, posing a risk to human health. Soil in urban areas deserves particular attention because it is often affected by human activities more than in non-urban areas and may in turn affect human health substantially. Soil is a natural source of antibiotics that have saved countless human lives. However, overuse of antibiotics in animal husbandry and in treating human illnesses has caused the spread of antibiotic resistance genes in the environment, raising concern about the evolution and spread of superbugs. Recognizing the important link of soil with human health, we have included four review articles in this special section. Steffan *et al.* (2018) present an overview of the effect of soil on human health, discussing the recent advances in understanding both the positive and negative, direct and indirect effects. They call for inter- and trans-disciplinary approaches in the research on the topic. Cakmak & Kutman (2018) discuss the conditions and distribution of zinc-deficient soils, leading to areas with small intakes of zinc by humans. Based on their vast experience of field trials, they describe cost-effective methods to biofortify cereal crops with zinc to tackle the global zinc malnutrition problem. Li *et al.* (2018) consider the importance of urban soil in maintaining human health,

focusing on pollutants and microbial biodiversity in urban soil and green spaces in urban areas. Xie *et al.* (2018) review the transfer of antibiotics and the spread of antibiotic resistance genes from animal manures to soil, and summarize the pathways, mechanisms and implications for human health of the dissemination of antibiotic resistance genes. Although the four articles deal with only a subset of the issues regarding soil and human health, it is hoped that they will serve as useful references and spur further research on this important topic.

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References

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